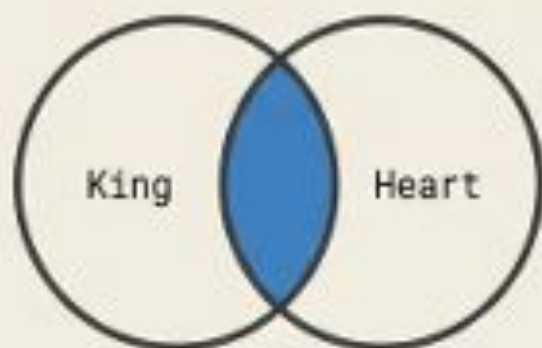


One Word, Two Very Different Meanings

THE ADDITION RULE



AND = At the Same Time

Context: Single Trial

Question: Is the card a King AND a Heart?

THE MULTIPLICATION RULE

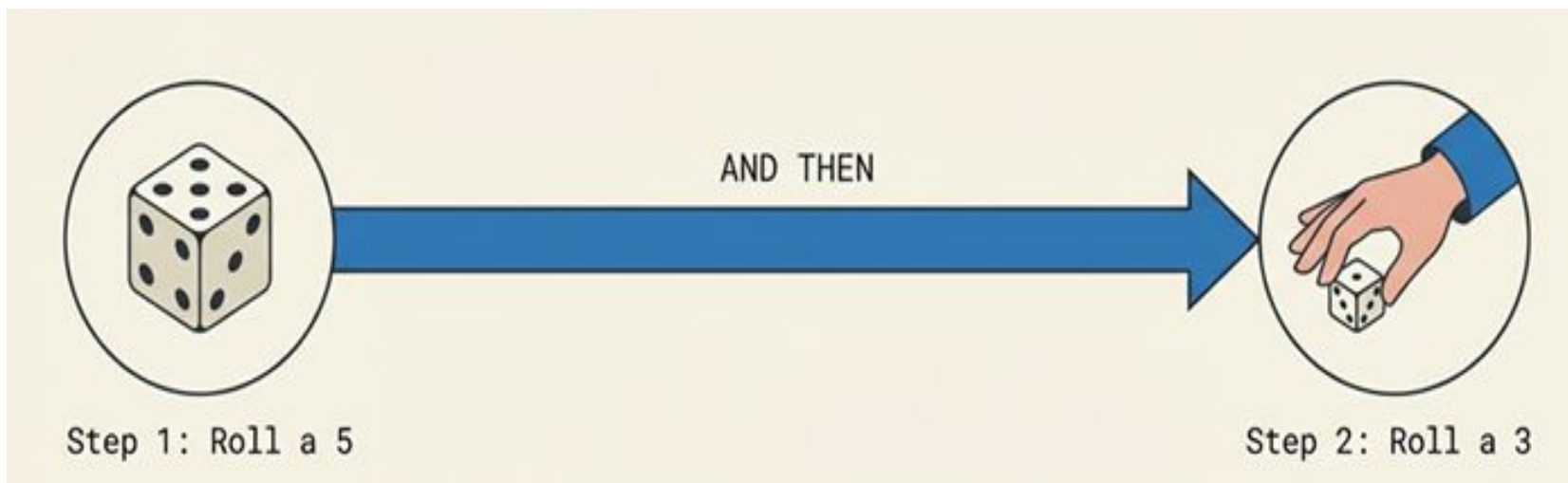


AND = Successive Events

Context: Multiple Trials

Question: Roll a 5, AND THEN roll a 3.

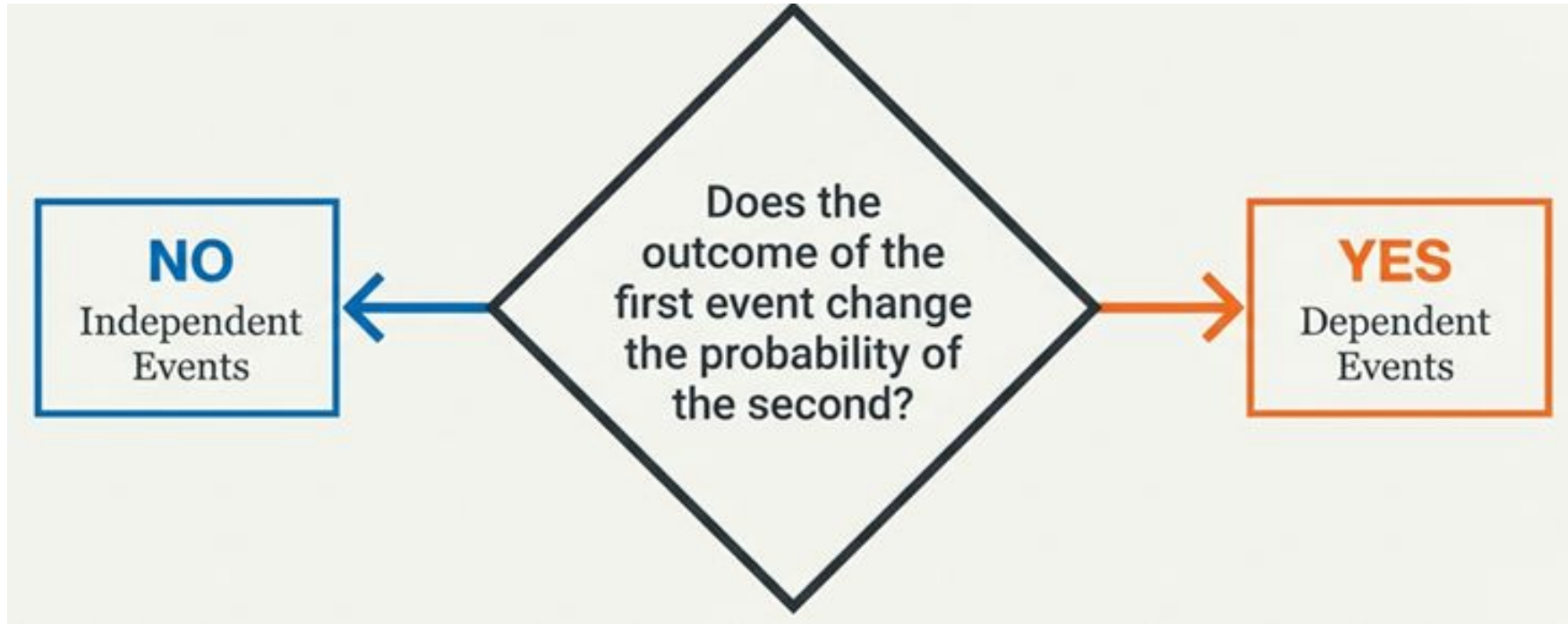
Key Takeaway: In multiplication problems, translate "And" to "**And Then**".



We are calculating the probability of the first event occurring, followed by the second event occurring.

$$P(A \text{ and } B) = P(A) \dots$$

Does Event A Change Event B?



Independent Events



Event A: The Coin Flip



Event B: The Die Roll

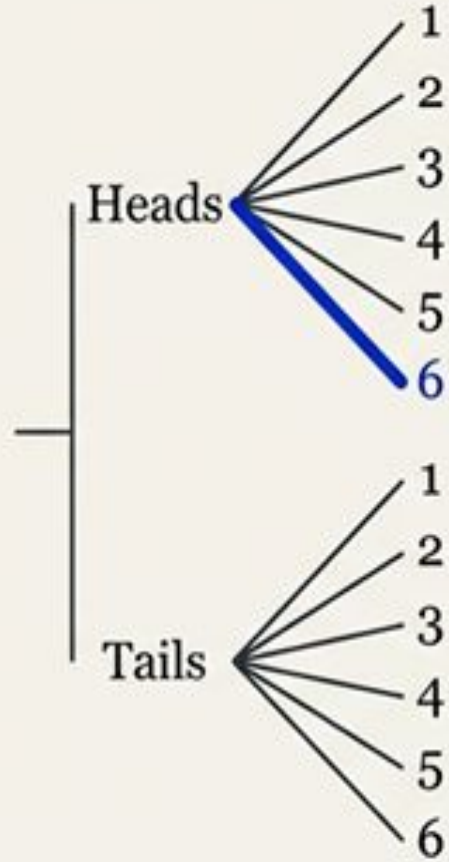
Definition: Events where the outcome of one has no effect on the probability outcome of the other. The coin has no memory of the die.

Rule #1: The Independent Formula

$$P(A \text{ and } B) = P(A) \times P(B)$$

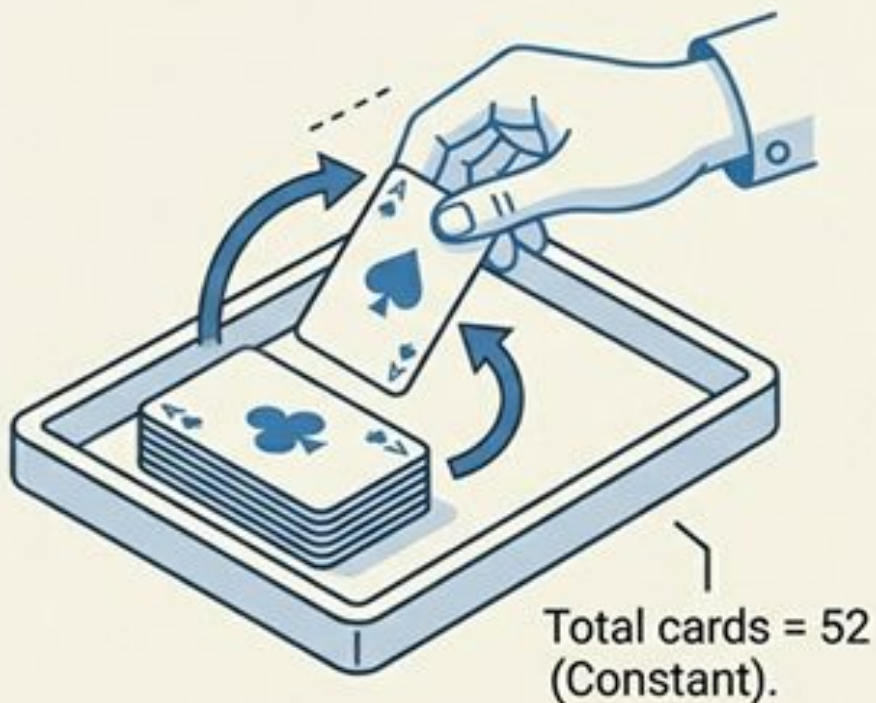
$$\frac{1}{2} \text{ (Coin)} \times \frac{1}{6} \text{ (Die)} = \frac{1}{12}$$

Probability = 8.3%

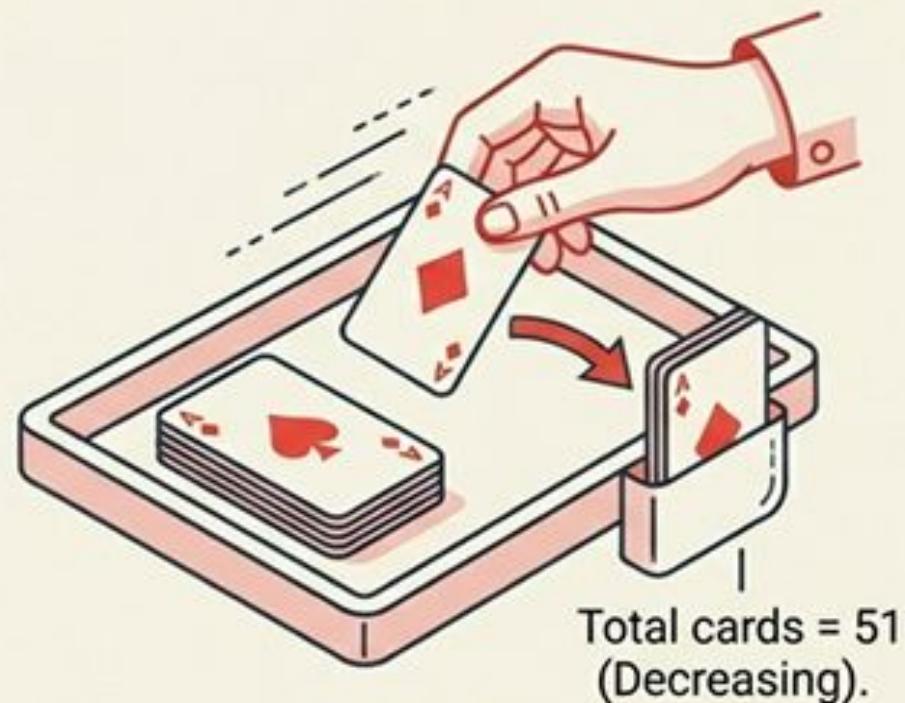


The "Replacement" Factor

With Replacement (Independent)



Without Replacement (Dependent)



"Without Replacement" is the specific phrase that signals a Dependent event.

Drawing Two Queens

Condition: Without Replacement

First Draw: $4/52$



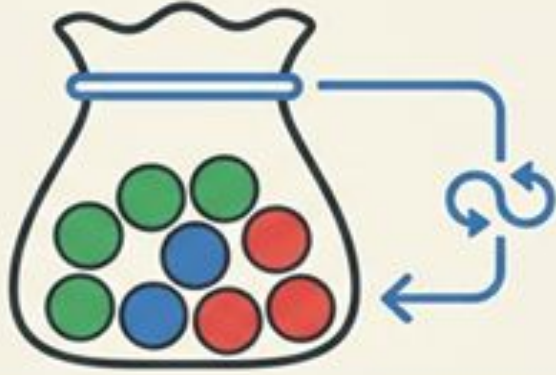
Second Draw: $3/51$



$$4/52 \times 3/51 = 12/2652 \approx 0.0045$$

In dependent events, you must account for exactly what was removed (both from the total and the specific group).

Case Study: The Green and Blue Marbles



Independent Loop

$$\frac{4}{9} \times \frac{2}{9} = \frac{8}{81}$$

First draw replaced, sample space remains 9.



Dependent Linear

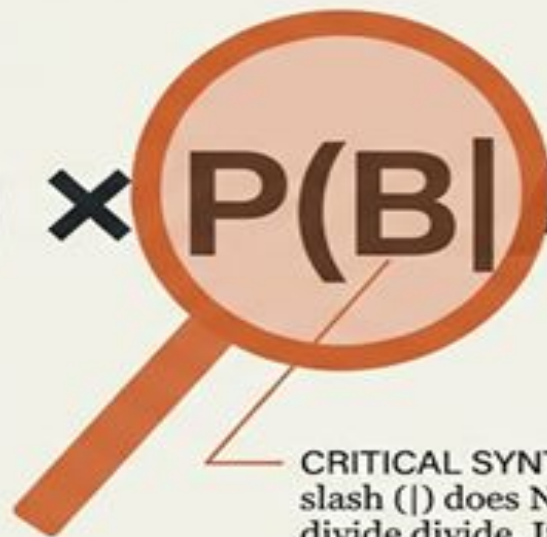
$$\frac{4}{9} \times \frac{2}{8} = \frac{8}{72} = \frac{1}{9}$$

First draw removed, sample space shrinks to 8.

Without replacement, the sample space shrinks, changing the probability of the second draw.

Rule #2: The Dependent Formula

$$P(A \text{ and } B) = P(A) \times P(B|A)$$



CRITICAL SYNTAX: The slash (/) does NOT mean divide. It reads: "The probability of B, Given that A has already occurred."

If, in an experiment, the events $A_1, A_2, \dots, A_k$ can occur, then

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_k) \\ = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_k|A_1 \cap A_2 \cap \dots \cap A_{k-1}). \end{aligned}$$

If the events $A_1, A_2, \dots, A_k$ are independent, then

$$P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1)P(A_2) \dots P(A_k).$$

Conditional probability

The conditional probability of B , given A , denoted by $P(B|A)$, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$

Two events A and B are **independent** if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A),$$

assuming the existences of the conditional probabilities. Otherwise, A and B are **dependent**.

Example:

The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$; the probability that it arrives on time is $P(A) = 0.82$; and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane

(a) arrives on time, given that it departed on time, and (b) departed on time, given that it has arrived on time.

Using Definition 2.10, we have the following.

(a) The probability that a plane arrives on time, given that it departed on time, is

$$P(A|D) = \frac{P(D \cap A)}{P(D)} = \frac{0.78}{0.83} = 0.94.$$

(b) The probability that a plane departed on time, given that it has arrived on time, is

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95.$$

c. The probability that it arrives on time, given that it did not depart on time.

$$P(A|D') = \frac{P(A \cap D')}{P(D')} = \frac{0.82 - 0.78}{0.17} = 0.24.$$

Example:

The concept of conditional probability has countless uses in both industrial and biomedical applications. Consider an industrial process in the textile industry in which strips of a particular type of cloth are being produced. These strips can be defective in two ways, length and nature of texture. For the case of the latter, the process of identification is very complicated. It is known from historical information on the process that 10% of strips fail the length test, 5% fail the texture test, and only 0.8% fail both tests. If a strip is selected randomly from the process and a quick measurement identifies it as failing the length test, what is the probability that it is texture defective?

Consider the events

L : length defective, T : texture defective.

Given that the strip is length defective, the probability that this strip is texture defective is given by

$$P(T|L) = \frac{P(T \cap L)}{P(L)} = \frac{0.008}{0.1} = 0.08.$$

Thus, knowing the conditional probability provides considerably more information than merely knowing $P(T)$.

Example 2.36: Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?

Solution: We shall let A be the event that the first fuse is defective and B the event that the second fuse is defective; then we interpret $A \cap B$ as the event that A occurs and then B occurs after A has occurred. The probability of first removing a defective fuse is $1/4$; then the probability of removing a second defective fuse from the remaining 4 is $4/19$. Hence,

$$P(A \cap B) = \left(\frac{1}{4}\right) \left(\frac{4}{19}\right) = \frac{1}{19}.$$



Example 2.37: One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?

Solution: Let B_1 , B_2 , and W_1 represent, respectively, the drawing of a black ball from bag 1, a black ball from bag 2, and a white ball from bag 1. We are interested in the union of the mutually exclusive events $B_1 \cap B_2$ and $W_1 \cap B_2$. The various possibilities and their probabilities are illustrated in Figure 2.8. Now

$$\begin{aligned}P[(B_1 \cap B_2) \text{ or } (W_1 \cap B_2)] &= P(B_1 \cap B_2) + P(W_1 \cap B_2) \\&= P(B_1)P(B_2|B_1) + P(W_1)P(B_2|W_1) \\&= \left(\frac{3}{7}\right) \left(\frac{6}{9}\right) + \left(\frac{4}{7}\right) \left(\frac{5}{9}\right) = \frac{38}{63}.\end{aligned}$$



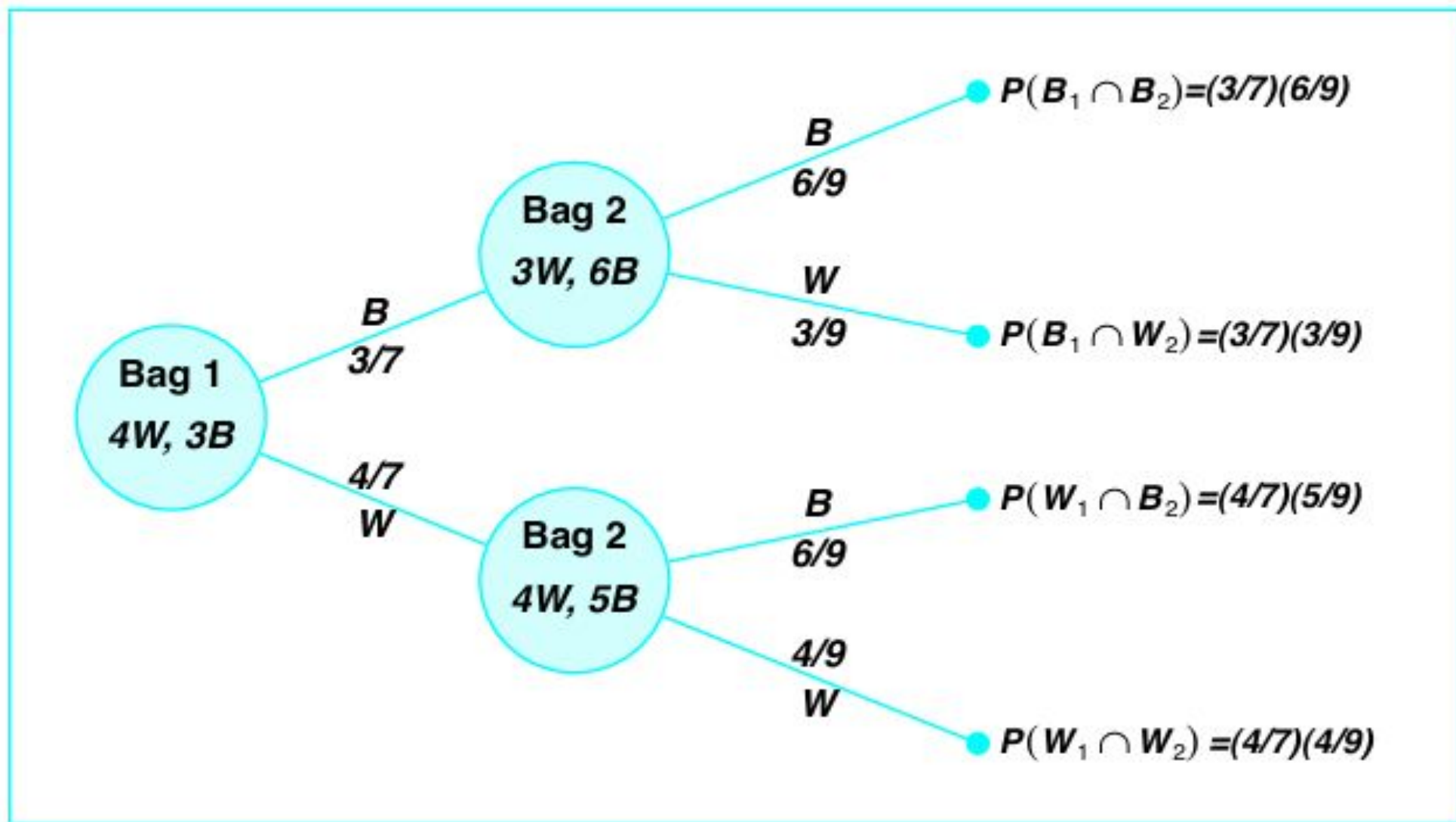


Figure 2.8: Tree diagram for Example 2.37.

Example 2.38: A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.

Solution: Let A and B represent the respective events that the fire engine and the ambulance are available. Then

$$P(A \cap B) = P(A)P(B) = (0.98)(0.92) = 0.9016.$$



Example:

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs, where A_1 is the event that the first card is a red ace, A_2 is the event that the second card is a 10 or a jack, and A_3 is the event that the third card is greater than 3 but less than 7.

First we define the events

A_1 : the first card is a red ace,

A_2 : the second card is a 10 or a jack,

A_3 : the third card is greater than 3 but less than 7.

Now

$$P(A_1) = \frac{2}{52}, \quad P(A_2|A_1) = \frac{8}{51}, \quad P(A_3|A_1 \cap A_2) = \frac{12}{50},$$

and hence, by Theorem 2.12,

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \\ &= \left(\frac{2}{52}\right) \left(\frac{8}{51}\right) \left(\frac{12}{50}\right) = \frac{8}{5525}. \end{aligned}$$

2.76 In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals:

	Nonsmokers	Moderate Smokers	Heavy Smokers
H	21	36	30
NH	48	26	19

where H and NH in the table stand for *Hypertension* and *Nonhypertension*, respectively. If one of these individuals is selected at random, find the probability that the person is

- (a) experiencing hypertension, given that the person is a heavy smoker;
- (b) a nonsmoker, given that the person is experiencing no hypertension.

2.87 A real estate agent has 8 master keys to open several new homes. Only 1 master key will open any given house. If 40% of these homes are usually left unlocked, what is the probability that the real estate agent can get into a specific home if the agent selects 3 master keys at random before leaving the office?